

```
errors = abs(ess.loc[:, 'predicted'] - ess.loc[:, 'happy'])  
  
print(np.mean(errors))
```

When we run this, we find that the mean error made by predictions in our decision tree is 1.369. This is higher than zero but lower than it might be if we used a worse prediction method. Our decision tree seems to make reasonably good predictions so far.

The Problem of Overfitting

You may have noticed one very important way that our method for evaluating our decision tree doesn't resemble how predictions work in real life. Remember what we did: we used the full set of survey respondents to generate our decision tree, and then we used that same set of respondents to judge the accuracy of our tree's predictions. But it's redundant to predict the happiness ratings of respondents who already took the survey—they took the survey, so we already know their happiness ratings and don't need to predict them at all! This would be like getting a dataset of past heart attack patients, meticulously studying their pretreatment symptoms, and building a machine learning model that told us whether someone had a heart attack last week. By now, it's already quite clear whether that person had a heart attack last week, and there are better ways to know than by looking at their initial triage diagnosis data. It's easy to predict the past, but remember that true prediction is always about the future. As Wharton professor Joseph Simmons put it, "History is about what happened. Science is about what happens *next*."

You may think that this isn't a serious problem. After all, if we can make a decision tree that works well with last week's heart attack patients, it's reasonable to suppose that it will work well with next week's heart attack patients. This is true to some extent. However, there is a danger that if we aren't careful, we can encounter a common, dastardly peril called *overfitting*, the tendency of machine learning models to achieve very low error rates on the datasets used to create them (like data from the past) and then unexpectedly high error rates on other data (like the data that actually matters, from the future).